

1. Introduction

The Low-Level Design (LLD) describes the internal working of the RAG-based Customer Support Assistant system. It focuses on module-level implementation, data structures, workflow logic, and system behavior.

2. Module-Level Design

1. Document Processing Module

- Loads the PDF using PyPDF
- Extracts text from each page
- Combines all text into a single string

2. Chunking Module

- Splits text into smaller chunks (size = 500 characters)
- Maintains overlap for better context understanding

3. Embedding Module

- Uses SentenceTransformer model
- Converts text chunks into numerical vectors
- Converts user query into vector format

4. Vector Storage Module

- Stores embeddings in memory using NumPy arrays
- Each chunk is associated with its embedding

5. Retrieval Module

- Computes similarity using dot product
- Retrieves top-k relevant chunks (k=3)
- Forms context from retrieved chunks

6. Query Processing Module

- Accepts user input
- Converts input into embedding
- Passes query to retrieval module

7. Graph Execution Module (LangGraph)

- Controls workflow using nodes
- Defines execution order of system

8. HITL Module

- Detects low-confidence queries
- Transfers control to human agent
- Accepts manual input as final response

3. Data Structures

text: str

Chunk Format

chunks: List[str]

Embedding Structure

`embeddings: numpy.ndarray`

Query-Response Schema

```
{  
  "query": str,  
  "answer": str,  
  "escalate": bool  
}
```

Graph State Object

```
class State(dict):  
    query: str  
    answer: str  
    escalate: bool
```

4. Workflow Design (LangGraph)

Nodes

- **Process Node** → Handles retrieval + LLM
- **Output Node** → Returns final result

Edges

Process Node → Output Node

5. Conditional Routing Logic

The system uses a condition to determine response handling:

Auto Response Condition

- If similarity score is high → generate answer

Escalation Condition

- If similarity score < threshold (e.g., 0.3)
- No relevant chunks found
- Query outside document scope

6. HITL Design

Trigger Condition

- Low similarity score
- No meaningful context

Process

1. System flags query as low confidence
2. Displays escalation message
3. Human agent provides response
4. System outputs human response

7. API / Interface Design

Input

query: str

Output

```
{  
  "answer": str,
```

```
"escalate": bool  
}
```

Interaction Flow

1. User inputs query
2. System processes via graph
3. Response returned or escalated

8. Error Handling

Case 1: Missing PDF

- System throws file not found error

Case 2: Empty Document

- No chunks generated

Case 3: No Relevant Chunks

- Trigger escalation

Case 4: Model Failure

- Return fallback message